The Limits to Growth: Sustainability and the Circular Economy

Lecture 0: Organization

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M.Sc. Anant Sujatanagarjuna
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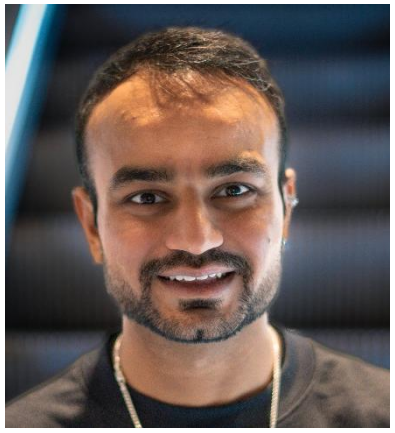
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- Updated versions of these slides will be available in our Github repository.

Team



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Research Group

- **E**merging **T**echnologies for the **C**ircular **E**conomy → **ETCE**
- Research focus:
  - Intersection of IT and sustainability
  - Circular Economy
  - Self-organized, decentralized and distributed systems
  - Machine-to-Everything Economy (M2X Economy)
- Other courses:
  - Requirements Engineering (WS – M.Sc.)
  - Emerging Technologies for the Circular Economy (SS – M.Sc.)

Research Group

- ETCE Website – Link
  - Course material
  - Theses/project topics

- Our research in action:
  - ZDF documentary (German) – Link
  - Klartext Preis 2020 (German) – Link

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You want join us? Write us an email!

→ benjamin.leiding@tu-clausthal.de

THE COURSE

Course Content

- Basics of climate change, environmental pollution, and dwindling non-renewable resources
- Introduction to the circular economy, sustainability, and related concepts (biocapacity, etc.)
- Sustainability goals
- Feedback loops and tipping points
- Implications of closed systems with a finite supply of resources
- Technology-focused and technology-critical approaches towards sustainability
- Circular Societies

Learning Outcome

- Understanding the concept of a circular economy, sustainability, and related concepts (biocapacity, etc.).
- Gain a basic understanding of causes, dimensions, and the characterization of climate change, environmental pollution, and dwindling non-renewable resources.
- Being able to make high-level, transdisciplinary assessments of decisions and measures in a social, economic, and political context.
- The ability to critically assess upcoming technological solutions enabling/facilitating sustainability and the circular economy.

Lectures

Date	Lecture Title
03.11.2026	L00 – Organisation & L01 – Introduction
10.11.2026	L02 – Challenges I – Environmental Pollution and Resource Scarcity
17.11.2026	L03 – Challenges II – Climate Change
24.11.2026	*L04 – Limits to Growth & Planetary Bounaries* ***(MOOC - Homework)***
01.12.2026	L05 - Circular Economy (CE)
08.12.2026	L06 – Sustainability and Political (In-)Action
15.12.2026	L07 – Circular Societies L08 – Complex Societies and Technology
22.12.2026	*L09 – Life Cycle Planning for Durable Products* ***(MOOC – Recording - Homework)***
05.01.2027	*L10 – Ethics / Moral Sustainability by Johannes Müller-Salo* ***(MOOC – Recording - Homework)***
12.01.2027	L11 – Beyond the Circular Economy
19.01.2027	L12 - LCA - Part 1
26.01.2027	L13 - LCA - Part 2
02.02.2027	Group Presentations and Lessons Learned
09.02.2027	L14 – Summary
16.02.2027	**Exam Q&A**

MOOC == Watch@Home

Exercises

Publication Date	Submissio Deadline	Exercise
03.11.2026	10.11.2026	E00 – Intro && E01 – Ecological Footprint
10.11.2026	17.11.2026	E02 – Household Waste
01.12.2026	08.12.2026	E03 – Der Verein (Group Exercise)
15.12.2026	22.12.2026	E04 – Protest Manifesto (Group Exercise)
22.12.2026	05.01.2027	E05 – Circular Societies (Group Exercise)
19.01.2027	02.02.2027	E06 – LCA (Group Exercise) && E07 – LCA Part 2 (Group Exercise)
02.02.2027		Group presentations

Course Organization

- Course website – Link

- News and updates:
  - All updates will be communicated through the university email
  - CLZ students + DigiTec will additionally receive information via StudIP (Link)

- Slides will be uploaded to Github (Link) → Please report bugs ;)

- Lecture recordings will be available on StudIP and on GitHub

- Questions? Write us an email: etce-ltg@tu-clausthal.de ← **We will only respond to emails written to this specific email address!**

Course Organization - Asynchronous Learning & MOOC content

- **M**assive **O**pen **O**nline **C**ourse
  - Remote and (often) asynchronous online courses *not* just for students enrolled in a specific university, but *ideally* open for everybody
  - Usually consist of pre-recorded lectures, interactive content and online quizzes
  - Some of you might have visited MOOC on platforms such as edX, LinkedIn Learning, Coursera, Udacity, etc. before

- We are currently developing a MOOC for the Limits to Growth Lecture

- This semester will be another test run for this asynchronous and digital learning content
  - We are very happy about any feedback you can give us to improve the course further! Just write us an email: etce-ltg@tu-clausthal.de

Course Organization – Asynchronous Learning

- This semester includes asynchronous learning content for some of the lectures
  - Consisting of short pre-recorded videos and interactive content
- You will get further information about these two sessions during the semester
  - You will find the lecture videos on the course website

24.11.2026	*L04 – Limits to Growth & Planetary Bounaries* ***(Flipped Classroom → MOOC@Home)***
22.12.2026	*L09 – Life Cycle Planning for Durable Products****(Flipped Classroom → MOOC@Home)***
05.01.2027	*L10 – Ethics/Moral Sustainability by Johannes Müller-Salo****(Flipped Classroom → MOOC@Home)***

The MOOC lectures will **not** be live lectures. Instead, you will find pre-recorded videos and other content on our website.

Dates/Times/Locations

- Lecture:
  - Tuesday **2:00 pm to 3:30 pm** (Berlin time) – 03.11.2026 to 16.02.2027
  - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton (Link)
- Exercise / Q&A:
  - Tuesday **3:45 pm to 4:40 pm** (Berlin time) – 03.11.2026 to 16.02.2027
  - Location: Goslar Gotec (Am Stollen 19 C, 38640 Goslar, Germany) or via BigBlueButton (Link)

EXERCISES

Exercises

- Individual work + group submissions
- Submission of **each** exercise is **mandatory**
- You pass by submitting an exercise with proper filename – even if it is an empty
- You will receive feedback on your submission
- Exercise = learning feedback

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- Exercise = learning feedback

Some exercises require you to submit your work. All such exercises should be submitted through the link in exercise sheet, using the password “limitstogrowth2627”:

- We do not accept email submissions; please use the file drop link mentioned in exercise sheet to upload your submissions.

- **Important**: Always submit the exercise sheet with correct filename for example,
  → *E01-Firstname.pdf*

Exercises

- Interactive PDF for each exercise.
- Answers to be filled in exercise PDF sheet itself.

- For Group submission(s) we strongly encourge you to use Q&A sessions to discuss and develop your ideas.

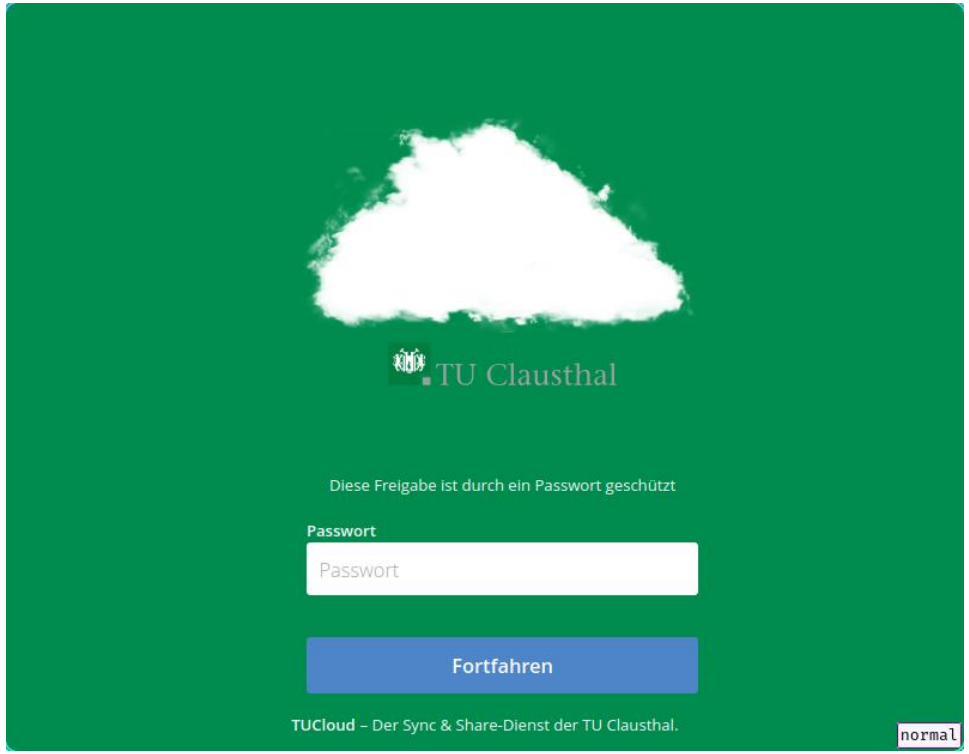
Exercises

Submission Instructions

- Step 1: Find the link in the exercise PDF files.

  Each exercise has a unique link to submission folder, which looks like this -->

- Step 2: Enter the password ‘limitstogrowth2627’



Exercises

Submission Instructions

- Step 3: Upload a file, e.g. “E01-Firstname.pdf”
  - Once you upload a file, you cannot delete it
  - Unless otherwise specified, we only accept PDF or mp4 files.
  - Ensure that your full name (Individual exercises) or Group name (Group exercise), and the exercise is mentioned in the filename.



Exercises

Submission Instructions

- Step 4: If your exercise was successfully uploaded,it will be visible. Please do not upload duplicates.



Bonus Exercises

- Simple and facile, sometimes also fun
- Not mandatory

- All student / Groups will have advantage in Exam for bonus task(s)
  - Submit all bonus task(s) to gain points
  - Eligible students will have a chance to skip **ONE** question in exam without adversly affecting final grading

EXAMINATION

Examination

- Prerequisite for admission to the final exam (all criteria have to be fulfilled):
  - Submit all exercises

- **Final exam:**
  - **Most likely on the 03.03.27 - 05.03.27**
  - Either written exam (120min) **or** oral examination (20-25min)

Examination

- Prerequisite for admission to the final exam (all criteria have to be fulfilled):
  - Submit all exercises

- **Final exam:**
  - **Most likely on the 03.03.27 - 05.03.27**
  - Either written exam (120min) **or** oral examination (20min)

- **Note for external students** → You do not have to register as a guest student at TU Clausthal! You pass the exam and get a written confirmation from us which you can present to your local examination office.

FINAL NOTES

Self-Study Star

Self-Study Star → 

- Slides with the self-study star indicate optional/additional study material that is **not** mandatory but could be helpful or interesting

Literature

- This course is not based on a single book and you **do not** need to buy a book to pass the exam.

- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *The Limits to Growth* (1972).
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. *Limits To Growth: The 30-Year Update* (2004).
- Baccini et al. *Metabolism of the Anthroposphere: Analysis, Evaluation, Design* (2012).
- Walter R. Stahel. *The Circular Economy: A User's Guide* (2019).
- W. Brian Arthur. *The Nature of Technology: What It Is and How it Evolves* (2011).
- David Wallace-Wells. *The Uninhabitable Earth, Annotated Edition* (2017).
- James Lawrence Powell. *The 2084 Report: An Oral History of the Great Warming* (2020).
- Joseph Tainter. *The Collapse of Complex Societies* (1988).

Literature

- (German) Stefan Rahmstorf, Hans Joachim Schellnhuber. *Der Klimawandel* (2019).
- David Archer, Stefan Rahmstorf. *The Climate Crisis* (2010).
- Gabrielle Walker, David King. *The Hot Topic: How to Tackle Global Warming and Still Keep the Lights on* (2008).

Questions?